Supplementary Materials

Contents

1 S0: Target Severity Distribution **2**

2 S1: Classification Method Details **3**
2.1 S1.1 Zero-Shot BART-MNLI . . . 3
2.2 S1.2 Confidence-Weighted Adjustment . . . 3
2.3 S1.3 Rule-Based Classifier . . . 3
2.3.1 S1.3.1 Weight Derivation Equation . . . 3
2.3.2 S1.3.2 Full Keyword Weights Table . . . 3
2.3.3 S1.3.3 Algorithm 1: Keyword Weight Derivation . . . 3
2.3.4 S1.3.4 Algorithm 2: Severity Classification . . . 4

3 S2: Evidence-Based Classifier **5**
3.1 S2.1 Validation Metrics Summary . . . 5
3.2 S2.2 Score Statistics by Severity Class (Test Set) . . . 5

4 S3: Statistical Analysis Details **6**
4.1 S3.1 McNemar's Test for Paired Classifier Comparison . . . 6
4.2 S3.2 Wilson Score Confidence Intervals . . . 6
4.3 S3.3 Jensen-Shannon Divergence . . . 6
4.4 S3.4 Chi-Square Goodness of Fit . . . 6
4.5 S3.5 Cohen's Kappa Significance Test . . . 6
4.6 S3.6 Cohen's Kappa Interpretation . . . 7

5 S4: Data and Code Availability **8**
5.1 S4.1 Input Data . . . 8
5.2 S4.2 Output Data . . . 8
5.3 S4.3 Reproducibility . . . 8

1 S0: Target Severity Distribution

Table 1: DDInter Evidence and Selected Targets

Severity	**DDInter (matched)**	**Selected Target**
Contraindicated	Major: 22.1%	**4%**
Major	(combined)	**18%**
Moderate	73.5%	**74%**
Minor	4.4%	**4%**

2 S1: Classification Method Details

2.1 S1.1 Zero-Shot BART-MNLI

$$\hat{y} = \arg\max_{c \in \mathcal{C}} P(\text{entailment} \mid \text{description}, c) \tag{1}$$

where $\mathcal{C} = \{\text{Contraindicated, Major, Moderate, Minor}\}$.

2.2 S1.2 Confidence-Weighted Adjustment

$$\text{Severity}_{\text{adj}} = \begin{cases} \text{downgrade by 1 level} & \text{if } \hat{y} \in \{\text{Contraindicated, Major}\} \wedge c < 0.65 \\ \hat{y} & \text{otherwise} \end{cases} \tag{2}$$

where $\hat{y}$ is the predicted severity and c is the model's confidence score.

2.3 S1.3 Rule-Based Classifier

2.3.1 S1.3.1 Weight Derivation Equation

$$w_k = \ln\left(\frac{P(k|\text{Major})}{P(k|\text{Moderate})}\right) \tag{3}$$

2.3.2 S1.3.2 Full Keyword Weights Table

Table 2: Empirically-Derived Keyword Weights (from training set, n=26,014)

Keyword	**Weight**	**LR**	**Major %**	**Mod %**
Positive weights (predictive of Major):				
prolongation	+1.45	4.25×	11.6%	2.7%
bleeding	+1.03	2.80×	25.9%	9.2%
hemorrhage	+0.63	1.88×	9.2%	5.0%
anticoagulant	+0.57	1.76×	3.3%	1.8%
risk / severity	+0.33	1.39×	49.7%	35.3%
increased	+0.20	1.22×	64.0%	52.0%
Negative weights (predictive of Moderate/Minor):				
decrease	−0.30	0.74×	25.7%	35.3%
serum	−0.31	0.73×	13.3%	18.5%
therapeutic	−1.20	0.30×	3.3%	11.8%
efficacy	−1.27	0.28×	3.8%	14.5%
antihypertensive	−1.49	0.23×	0.5%	2.3%
reduce	−1.61	0.20×	0.3%	1.7%

LR = Likelihood Ratio = $P(k|\text{Major})/P(k|\text{Moderate})$. Weight = ln(LR).

2.3.3 S1.3.3 Algorithm 1: Keyword Weight Derivation

Input: $\mathcal{D}_{\text{train}}$ (training set), $\mathcal{K}$ (candidate keywords)
Output: $\mathcal{W}$ (keyword → weight dictionary)

1. Count Major and Moderate instances: n_{Major}, n_{Moderate}
2. For each keyword $k \in \mathcal{K}$:
   - Count occurrences: c_{Major}, c_{Moderate}

- If $c_{\text{Major}} + c_{\text{Moderate}} \geq 50$:
  - $P(k|\text{Major}) \leftarrow c_{\text{Major}}/n_{\text{Major}}$
  - $P(k|\text{Moderate}) \leftarrow c_{\text{Moderate}}/n_{\text{Moderate}}$
  - $\mathcal{W}[k] \leftarrow \ln(P(k|\text{Major})/P(k|\text{Moderate}))$

3. Return $\mathcal{W}$

2.3.4 S1.3.4 Algorithm 2: Severity Classification

Input: $\mathcal{W}$ (keyword weights), $\mathcal{S}_{\text{train}}$ (training scores), d (DDI description)
Output: severity $\in$ {Contraindicated, Major, Moderate, Minor}
Step 1: Compute severity score

$$S = \sum_{k \in \mathcal{W}} \mathcal{W}[k] \cdot \mathbb{1}[k \in d] \tag{4}$$

Step 2: Compute percentile thresholds from training scores (grid-search optimized)

- $P_{96} \leftarrow \text{percentile}(\mathcal{S}_{\text{train}}, 96)$ (top 4%)
- $P_{92} \leftarrow \text{percentile}(\mathcal{S}_{\text{train}}, 92)$ (top 8%, optimized from 78%)
- $P_2 \leftarrow \text{percentile}(\mathcal{S}_{\text{train}}, 2)$ (bottom 2%, optimized from 4%)

Step 3: Classify based on score

$$\text{severity} = \begin{cases} \text{Contraindicated} & \text{if } S \geq P_{96} \\ \text{Major} & \text{if } P_{92} \leq S < P_{96} \\ \text{Minor} & \text{if } S \leq P_2 \\ \text{Moderate} & \text{otherwise} \end{cases} \tag{5}$$

3 S2: Evidence-Based Classifier

Uses identical keyword weights as Rule-Based (Table 2) with an additional confidence penalty:

$$S_{\text{adjusted}} = \begin{cases} S_{\text{raw}} \times 0.8 & \text{if confidence} < 0.5 \\ S_{\text{raw}} & \text{otherwise} \end{cases} \tag{6}$$

Note: Achieves identical performance (66.4% exact accuracy) to Rule-Based because confidence penalization has minimal effect on the empirically-weighted scores.

3.1 S2.1 Validation Metrics Summary

Table 3: DDInter Validation Results (test set n=11,150)

Method	**Exact Acc. (95% CI)**	**Adjacent Acc.**	**Cohen's κ**
Zero-Shot BART-MNLI	22.1% (21.3–22.9%)	95.6%	0.000
Confidence-Weighted	22.1% (21.3–22.9%)	95.6%	0.000
Rule-Based (Optimized)	**66.4% (65.5–67.3%)**	**99.3%**	**+0.096**†
Evidence-Based	66.4% (65.5–67.3%)	99.4%	+0.096†

McNemar's test (Rule-Based vs. Zero-Shot): χ^2=2750.4, p <2.2×10^{-16}

†Cohen's κ significantly >0 (z=4.77, p <0.0001)

Improvement: +44.3 percentage points over zero-shot baseline (3× better, McNemar's p <2.2×10^{-16}).

Threshold Optimization: Grid search on training set (n=26,014) identified optimal thresholds: Major ≥ 92nd percentile, Minor ≤ 2nd percentile. Original thresholds (Major ≥ 78%, Minor ≤ 4%) achieved 52.1%; optimized thresholds achieve 66.4%.

3.2 S2.2 Score Statistics by Severity Class (Test Set)

Table 4: Keyword Score Discrimination by DDInter Ground Truth

Severity	**Mean Score**	**Std Dev**	**n**	**% of Test**
Major	+0.926	1.328	2,464	22.1%
Moderate	+0.065	1.477	8,193	73.5%
Minor	−0.281	1.250	493	4.4%

Interpretation: Mean scores show monotonic trend (Major > Moderate > Minor), confirming keyword weights discriminate severity. High within-class variance (std ≈ 1.3–1.5) limits exact classification accuracy.

4 S3: Statistical Analysis Details

4.1 S3.1 McNemar's Test for Paired Classifier Comparison

McNemar's test compares two classifiers on the same test set by constructing a 2×2 contingency table:

Table 5: McNemar's Contingency Table (Rule-Based vs. Zero-Shot)

	Zero-Shot Correct	**Zero-Shot Wrong**
Rule-Based Correct	502	6,900
Rule-Based Wrong	1,962	1,786

$$\chi^2 = \frac{(|b-c|-1)^2}{b+c} = \frac{(|1962-6900|-1)^2}{8862} = 2750.4 \tag{7}$$

where b = Zero-Shot only correct (1,962) and c = Rule-Based only correct (6,900). Result: $\chi^2 = 2750.4$, df=1, $p < 2.2 \times 10^{-16}$.

4.2 S3.2 Wilson Score Confidence Intervals

For accuracy proportions, we use the Wilson score interval (more accurate than normal approximation for bounded proportions):

$$\text{CI} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \tag{8}$$

where $\hat{p}$ is observed accuracy, $n = 11{,}150$, and $z = 1.96$ for 95% CI.
Results:

- Zero-Shot: 22.1% (95% CI: 21.3%–22.9%)
- Rule-Based: 66.4% (95% CI: 65.5%–67.3%)

4.3 S3.3 Jensen-Shannon Divergence

$$D_{JS}(P\|Q) = \frac{1}{2}D_{KL}(P\|M) + \frac{1}{2}D_{KL}(Q\|M) \tag{9}$$

where $M = \frac{1}{2}(P+Q)$.

4.4 S3.4 Chi-Square Goodness of Fit

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} \tag{10}$$

4.5 S3.5 Cohen's Kappa Significance Test

To test whether κ is significantly different from zero:

$$z = \frac{\kappa}{\text{SE}_\kappa}, \quad \text{where } \text{SE}_\kappa \approx \sqrt{\frac{p_e}{n(1-p_e)^2}} \tag{11}$$

For our Rule-Based classifier: $\kappa = 0.096$, SE ≈ 0.020, $z = 4.77$, $p < 0.0001$.

Table 6: Cohen's Kappa Interpretation Guide

Kappa Range	**Agreement Level**
< 0.20	Poor
0.21–0.40	Fair
0.41–0.60	Moderate
0.61–0.80	Substantial
0.81–1.00	Almost perfect

4.6 S3.6 Cohen's Kappa Interpretation

Our achieved $\kappa = +0.096$ is statistically significant ($z = 4.77$, $p < 0.0001$), indicating agreement better than chance. The low absolute value falls in the "poor" range, which is expected given DDInter's severe class imbalance (73.5% Moderate). The improvement from baseline ($\kappa = 0.000$) demonstrates meaningful classifier calibration.

5 S4: Data and Code Availability

5.1 S4.1 Input Data

- **Source**: DrugBank v5.1.9 (Creative Commons license)
- **Filter**: Cardiovascular and antithrombotic ATC codes
- **File**: `data/ddi_cardio_or_antithrombotic_labeled.csv`

5.2 S4.2 Output Data

- **Recalibrated predictions**: `data/ddi_semantic_final.csv`
- **Statistics**: `data/ddi_semantic_final.json`

5.3 S4.3 Reproducibility

```
pip install pandas numpy scipy matplotlib torch sentence-transformers
python recalibrate_severity_gpu.py
python publication_recalibration/generate_figures_updated.py
```